Executive Summary

1. Project Overview

- **Project Title:** Hospital Performance Analytics Dashboard
- **Client:** Apollo Hospitals (simulated)
- **Project ID:** IA4S-PBI-H001
- **Industry:** Healthcare & Hospital Operations
- **Duration:** 3 Days
- **Role:** Junior Data Analyst
- **Project Type:** End-to-End Power BI Capstone Project

Project Description This project simulates a real-world BI engagement where you design dashboards for hospital operations: bed management, patient analysis, doctor performance, and revenue trends

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Business Requirements

Bed Management

- Total Beds, Available Beds, Occupancy %, Beds under Maintenance
- Bed availability by branch, ward, state

Doctor & Patient Analysis

- Total Doctors, High-Rated Doctors, Avg Consult Fee
- Total Patients, Avg Patient Age, Gender % split
- Doctor ratings by department
- Admissions trend

Financial Analysis

- Profit Growth % (▲ / ▼ with color codes)
- Revenue by Patient Name (Top contributors highlighted)
- Avg Revenue per Department

City & Branch Analysis

- Occupancy % by city
- Beds by state
- Top vs Bottom performing branches

Dataset Collection

☐ **Source:** Kaggle hospital datasets (Admissions, Doctors, Patients, Beds).

☐ **Tables Required:**

- **FactAdmissions:** Patient visits, consult fees, bed usage.
- **DimDoctor:** Doctor details, ratings, department.
- **DimPatient:** Patient demographics, age, gender, city.
- **DimHospital:** Branch, ward, bed type, bed status.
- **DimDate:** Calendar table for time intelligence

Project Workflow

Business Requirements → Dataset Collection → Power Query Cleaning → Data Modeling → DAX KPIs → Dashboard Design → Insights → GitHub Publication → Final Submission

Dashboards (Each Page)

Page 1 – Bed Management Dashboard

- KPIs: Total Beds, Available Beds, Occupancy %, Beds under Maintenance
- Visuals: Bar chart (Beds by State), Table (Branch & Ward bed status), Map (Hospital locations)
- Insight: Bengaluru has highest occupancy; Vizag underutilized.

Page 2 – Patient & Doctor Overview Dashboard

- KPIs: Total Doctors, Total Patients, Avg Consult Fee, Avg Patient Age, High-Rated Doctors
- Visuals: Pie chart (Gender split), Bar chart (Doctor ratings), Horizontal bar chart (Doctors by Department), Line chart (Admissions trend), Table (Doctor details)
- Insight: Admissions rising steadily; high-rated doctors drive satisfaction.

Page 3 – Occupancy & City Analysis Dashboard

- KPIs: Occupancy % by city
- Visuals: Pie chart (City-wise occupancy %), Bar chart (Total vs Available Beds by state)
- Insight: Bengaluru leads occupancy; Vizag underutilized.

Page 4 – Financial & Revenue Dashboard

- KPIs: Profit Growth % (▲ / ▼ with color codes), Revenue by Patient Name, Avg Revenue per Department
- Visuals: KPI cards, Bar chart (Revenue by Patient Name), Table (Patient contribution %)
- Insight: Top 5 patients contribute majority of revenue; profit growth fluctuates month-to-month.

Page 5 – Advanced Analytics Dashboard (Optional Kaggle Dataset)

- KPIs: Patient Readmission Rate, Avg Treatment Cost per Department, Mortality Rate %
- Visuals: Line chart (Readmission trend), Bar chart (Treatment cost by department), Table (Mortality rate by branch)
- Insight: Certain departments have higher readmission rates.

Business Insights

- ☐ Bengaluru branch has highest occupancy (26%).
- ☐ Avg patient age ~45 years, balanced gender distribution (49% Female, 51% Male).
- ☐ High-rated doctors (136/200) drive patient satisfaction.
- ☐ Profit Growth % shows ▲/▼ trends month-to-month with color coding.
- ☐ Top 5 patients contribute majority of revenue

Recommendations

☐ Increase bed availability in high-demand cities (Bengaluru, Hyderabad).

☐ Focus on departments with lower ratings (Dermatology, Gastroenterology).

☐ Optimize consult fees for affordability vs revenue.

☐ Monitor patient repeat visits to improve loyalty.

☐ Expand capacity in underutilized branches (Vizag).

☐ Track YoY Profit Growth % for efficiency improvements